

THE CLAUDE BEAST

Portless: Named URLs for Local Dev

Replace Port Numbers with Stable .localhost URLs. For Humans and Agents.

Vercel Labs | 2,400+ Stars in 10 Days | Apache-2.0

The Port Number Problem

Nine pain points that multiply with every service in your stack



EADDRINUSE

Port conflicts when two projects use the same default port



Memorization Tax

Cognitive overhead remembering which service is on which port



Stale Tab Confusion

Stopping one server, starting another on the same port



Monorepo Scaling

Every problem above multiplies with each service in the repo



AI Agent Failures

Coding agents guess or hardcode wrong ports in tests



Cookie Bleeding

Cookies on localhost shared across all ports and apps



Config Brittleness

CORS, OAuth, .env files with hardcoded ports that break



Team Overhead

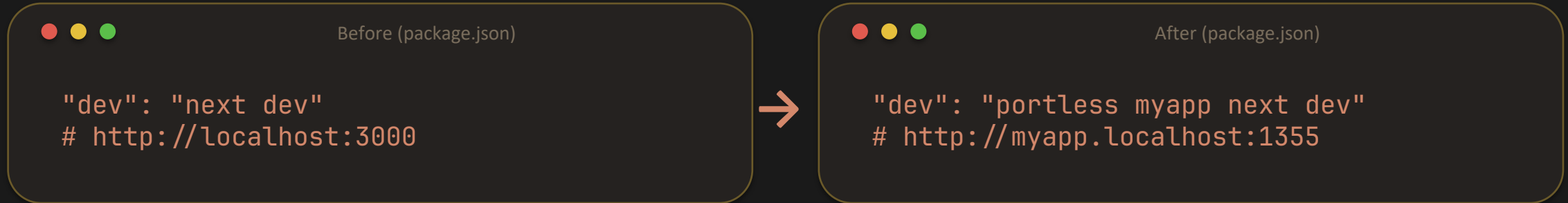
"What port is that on?" asks in Slack for every new dev



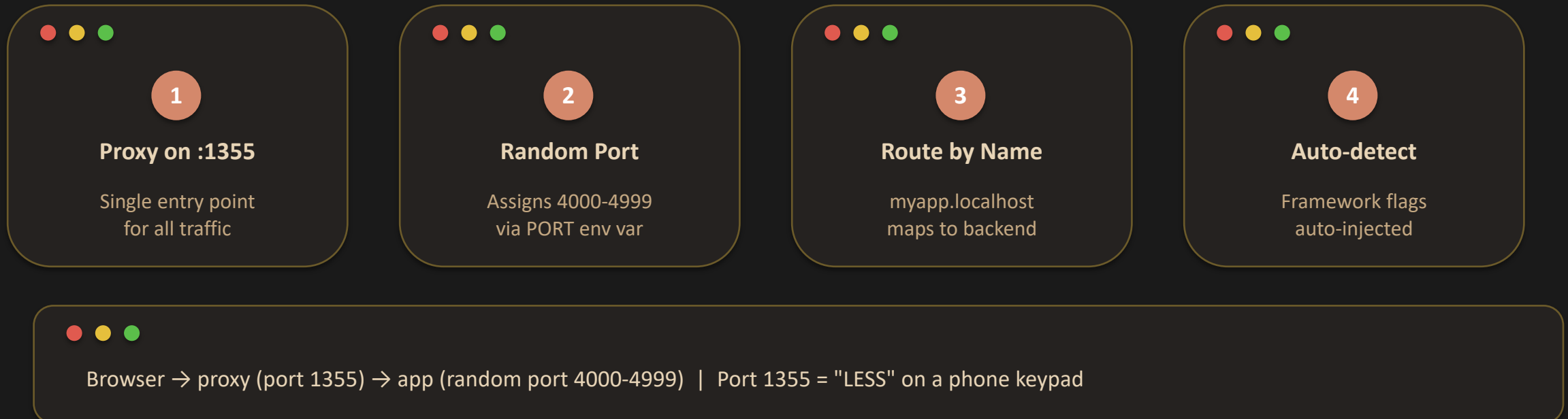
Useless History

Browser history for localhost:3000 is a jumble of projects

The Solution: Named .localhost URLs



How It Works



Why AI Agents Need Portless

"For humans and agents" -- the tagline is not an afterthought

The Agent Problem

- Agents frequently guess wrong ports
- Hardcoded ports break in monorepos
- Port changes between sessions confuse agents
- No semantic meaning: localhost:3001 tells nothing
- Test scripts with wrong ports give false failures

The Portless Solution

- Stable URLs: api.myapp.localhost:1355 always works
- Semantic naming: URL tells the agent what it is
- No port guessing or config file parsing needed
- Same URL across sessions, team members, agents
- Test scripts with stable URLs never break

Agents That Benefit

Claude Code

Stable endpoints for browser automation and testing

GitHub Copilot

Reliable URLs in agent mode for local verification

OpenAI Codex

Consistent targets for generated test scripts

Cursor / Windsurf

Named URLs for preview and integration testing

Technical Architecture

Request Flow



State Management

Ports < 1024: /tmp/portless (shared)
Ports >= 1024: ~/.portless (user-scoped)



Process Management

PID file tracking, orphan detection,
signal handling, terminal cleanup



Security

SHA-256 enforced for all certs,
per-hostname SANs, local CA



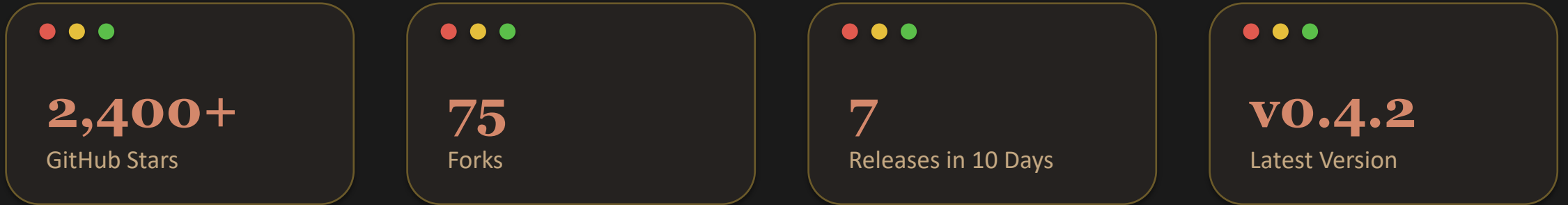
Shell Compatibility

Commands via /bin/sh -c
nvm, fnm, mise supported

TypeScript 72% | JavaScript 14% | Python 8% | pnpm monorepo | Turborepo

Adoption: 2,400+ Stars in 10 Days

Exceptional traction for a developer CLI tool | February 15 - 25, 2026



Release Velocity

v0.1.0	Feb 18	Initial release, basic proxy
v0.2.0	Feb 18	Custom proxy port
v0.3.0	Feb 18	Auto-start proxy, file-based routing
v0.4.0	Feb 18	HTTP/2, TLS certs, local CA
v0.4.1	Feb 22	E2E tests for 11 frameworks
v0.4.2	Feb 25	Shell compat, loop detection, SHA-256

Use Cases: Beyond Port Management



Monorepo Service Mesh

1

Frontend, API, docs, admin each get named URLs. Subdomains organize hierarchically.
api.myapp.localhost:1355, docs.myapp.localhost:1355



OAuth & CORS Stability

2

Redirect URLs and CORS allowlists use stable named URLs instead of brittle port numbers.
No more broken OAuth flows when ports shift.



Cookie Isolation

3

Cookies on localhost bleed across all ports. Named subdomains isolate cookies per-app.
No more auth state leaking between projects.



AI Agent Testing

4

Claude Code, Copilot, Codex all target stable semantic URLs instead of guessing ports.
Test scripts never break due to port changes.

Why Portless Matters

The Port Problem Is Real

Nine distinct pain points that multiply with every service. Port management is a hidden productivity tax on every developer, every day.

Agents Need Stable Endpoints

As AI coding agents become standard, they need semantic, reliable URLs for local services. Portless gives them exactly that.

Zero-Friction Adoption

`npm install -g portless`. One line change in `package.json`. 11 frameworks supported. HTTPS with one command.

github.com/vercel-labs/portless | [Apache-2.0](#) | port1355.dev